LIVE MEETING SUMMARIZER APPLICATION

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PROBLEM STATEMENT

- Manual note-taking in meetings is time-consuming and often misses important information like key decisions and action items.
- There is no efficient way to track speaker-wise contributions, making it difficult to understand discussions clearly.
- Existing systems lack an integrated solution for real-time speech-to-text, speaker diarization, and automated summarization.

PROPOSED SOLUTION

- Build a full-stack AI-based application that captures live or uploaded meeting audio and converts it into text using Speech-to-Text models.
- Integrate speaker diarization and LLM-based summarization to generate structured summaries with speaker identification, key decisions and action items.
- Provide a user-friendly interface with storage, download and sharing features for easy access and management of meeting summaries.

TECH STACK

- Frontend: React.js / Vite, CSS
- Backend: Python, FastAPI
- Database: MongoDB Atlas

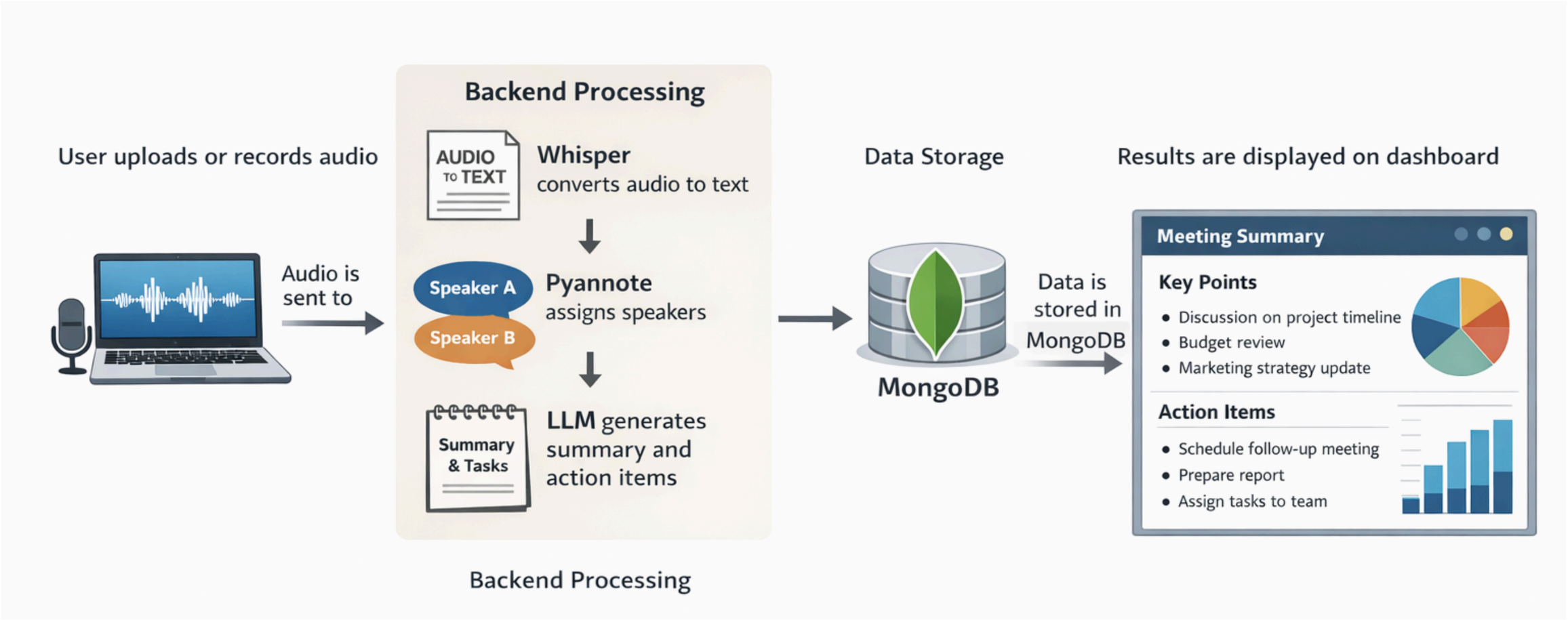
AI Models:

- Whisper (Speech-to-Text)
- Pyannote (Diarization)
- LLaMA 3 via Groq API (Summarization)

Deployment:

- Vercel (Frontend)
- Hugging Face Spaces (Backend)

WORKFLOW


User uploads or records audio
Audio is sent to
Backend Processing
AUDIO TO TEXT
Whisper converts audio to text
Speaker A
Speaker B
Pyannote assigns speakers
Summary & Tasks
LLM generates summary and action items
Backend Processing
Data Storage
MongoDB
Data is stored in MongoDB
Results are displayed on dashboard
Meeting Summary
Key Points
Discussion on project timeline
Budget review
Marketing strategy update
Action Items
Schedule follow-up meeting
Prepare report
Assign tasks to team

FEASIBILITY & VIABILITY

Feasibility:

- Uses lightweight STT + fast LLM → efficient performance without heavy compute

Viability:

- Cost-effective using free cloud platforms (Vercel, Hugging Face)

IMPACT AND BENEFITS

- Improves productivity by eliminating manual note-taking
- Ensures accountability with speaker-wise tracking
- Provides searchable history of meetings
- Reduces time spent reviewing long recordings by delivering concise summaries
- Enhances decision-making by clearly highlighting key points and action items
- Makes meeting information easily accessible and shareable across teams

CONCLUSION

- Transforms unstructured meeting audio into clear, structured, and actionable insights
- Demonstrates seamless integration of Speech-to-Text, speaker diarization, and LLM-based summarization
- Enhances productivity, collaboration, and decision-making in real-world scenarios
- Provides a scalable and extensible AI-driven solution for modern meeting environments

Live Website: https://live-meeting-summarizer-application.vercel.app

THANK YOU